



CODECHEF

The CodeChef.com Challenge, your monthly dose of coding puzzles from India's biggest coding contest, is now in print!

*H*ello and welcome to the *CodeChef Puzzle Challenge* sponsored by *LINUX For You* and *Direct!* Solve this month's puzzle, and get a chance to be one of the three lucky people to win a cash prize of Rs 1,000 each!

The March edition's puzzle:

Ram and Shyam sell donuts in the market. Ram sells jumbo donuts at two for Rs 50, while Shyam sells normal donuts at three for Rs 50. At a particular moment when both have the same amount of donuts left, Shyam is called away. He asks Ram to take care of his donuts. To prevent everything from getting too complicated, Ram puts all the donuts in one basket, and starts selling five donuts for Rs 100. When Shyam returns at the end of the day, all the donuts have been sold. But when they start dividing the money, there appears to be a shortage of Rs 350. Suppose they divide the amount of money equally between themselves, how much does Ram lose in this deal?

The solution:

Let 'y' be the number of donuts left with Ram and Shyam, respectively, when Shyam left. Hence,
Total number of donuts = 2y
Ram sold 5 donuts for Rs 100. Hence, the selling price of 2y donuts = $(2y/5) * 100 = 40y$ (1)
Ram would have sold his y donuts in $(y/2) * 50$, i.e. $25 * y$ (2)
Shyam would have sold his y donuts in $(y/3) * 50$.
So total selling price would have been $y * 25 + (y/3) * 50$ (3)
Now $(3) - (1) = 350$ (given)

i.e.: $5y = 1050$ (4)
...using (1), Ram's share = $(eqn 1)/2 = (40y)/2 = 20y$
but Ram would have earned 25y (using (2)). Therefore, Ram gets 5y less.
Using (3), $5y = 1050$, hence Ram lost Rs 1050 with this deal.

And the winners this month are:

- Prabha Rajagopalan
- Saurabh Modi
- Himanshu Sachan

Here's Your CodeChef Puzzle of the Month:

Consider two stations A and B located 100 kilometres apart. There is a station C, located between A and B. Now trains from station A and B start moving towards station C at different speeds. The train starting from A moves at 40 km/h while the train starting from B moves at 60 km/h. These trains have to stop at station C. A bird also starts from station B and flies at a speed of 80 km/h towards station A. When it reaches the train coming from station A, it switches direction – flying back towards station B and when it reaches the train coming from station B, it turns around again, heading towards the direction of station A. This goes on for some time till both the trains reach station C.
What is the total distance covered by the bird in the time it takes for the two trains to reach station C?

Looking for a little extra cash?

Solve this month's *CodeChef Challenge* puzzle and send in your answer to codechef@afyindia.com or lly@codechef.com by April 15, 2012. Three lucky winners get to win some awesome prizes both in cash and merchandise! You can also participate in the contest by visiting the online space for the article at <https://www.facebook.com/LinuxForYou> 

About CodeChef: CodeChef.com is India's first, non-commercial, online programming competition, featuring monthly contests in more than 35 different programming languages. Log on to CodeChef.com for the April Challenge that takes place from the 1st to the 11th, to win cash prizes up to Rs 20,000. Apart from that, keep visiting CodeChef and contests in the multiple programming contests that take place on the website throughout the month!